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EXPERIMENT 8

Aim: Configuring a Cryptocurrency Wallet using Metamask and integrating it with a DApp

THEORY

1. Cryptocurrency Wallet

A cryptocurrency wallet is a digital application used to manage blockchain accounts. It stores private keys that allow users to access their cryptocurrency and authorize transactions.

2. MetaMask

MetaMask is a cryptocurrency wallet and browser extension that allows users to interact with Ethereum networks and decentralized applications (DApps). It manages accounts, displays balances and signs transactions.

3. Ethereum Sepolia Test Network

Sepolia is an Ethereum test network used for testing blockchain applications and transactions without spending real ETH. Test ETH can be obtained through faucets.

4. Ethereum Wallet Address

A wallet address is a unique identifier of an Ethereum account. It is used to send and receive ETH and interact with smart contracts.

5. Decentralized Application (DApp)

A DApp is an application that interacts with blockchain technology. It uses a frontend interface, a Web3 provider and a blockchain network to perform decentralized operations.

6. Wallet–DApp Integration

Wallet–DApp integration allows users to connect their MetaMask wallet to a web application. After permission is granted, the DApp can access the wallet address and request blockchain transactions.

7. Ethers.js

Ethers.js is a JavaScript library used to connect web applications with Ethereum. It provides functions to access wallet accounts, retrieve balances and send transactions.

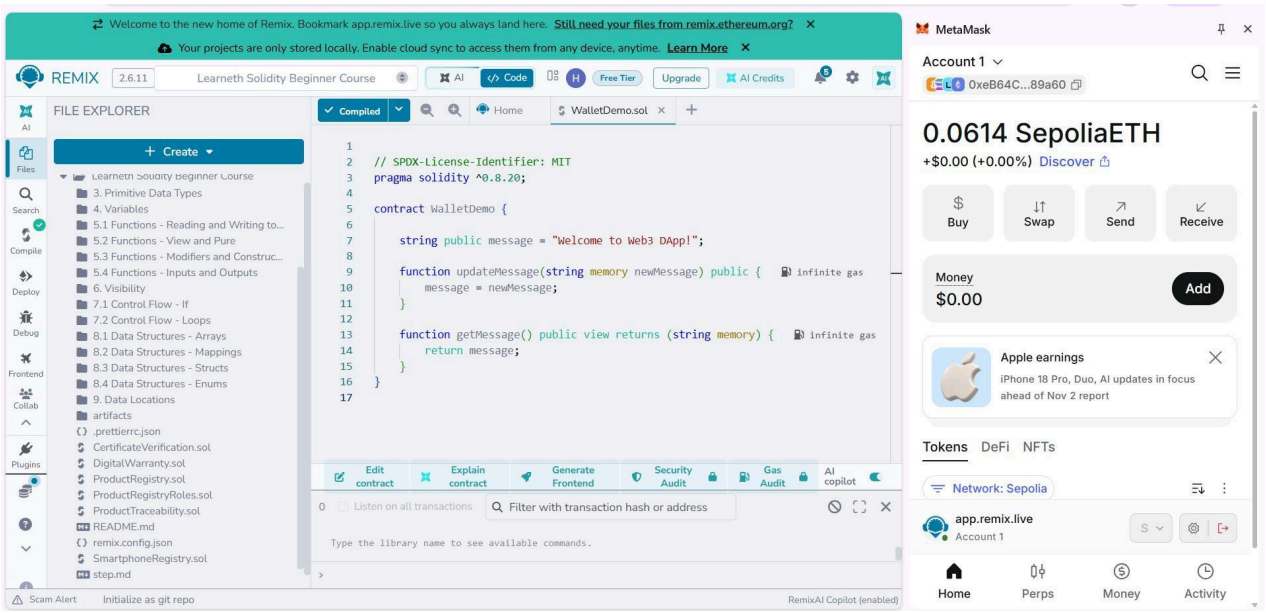
8. Gas Fees and Transactions

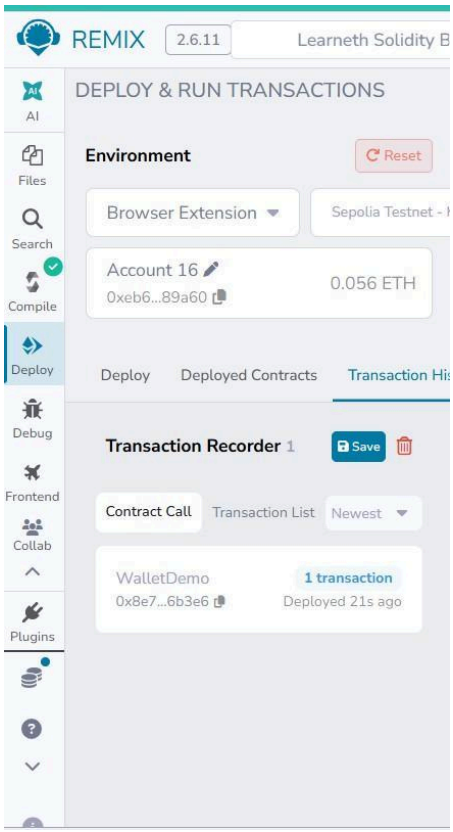
Gas fees are the transaction costs required to perform operations on the Ethereum network. MetaMask allows users to review and approve transactions before they are submitted to the blockchain.

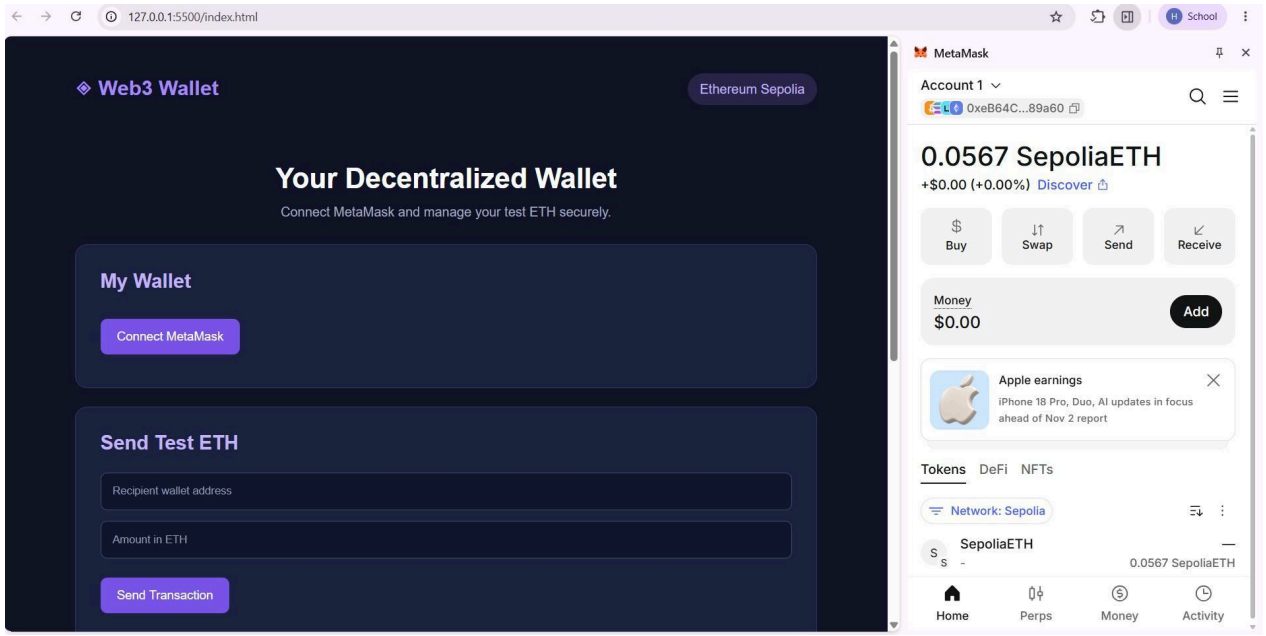
9. Blockchain Transaction Verification

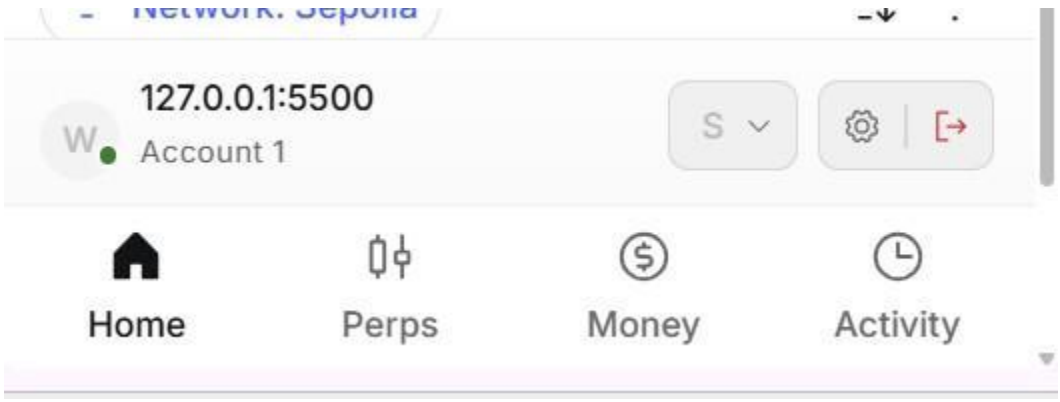
Transactions submitted to Ethereum are recorded on the blockchain. Sepolia Etherscan can be used to verify transaction details such as sender, receiver, amount and transaction status.

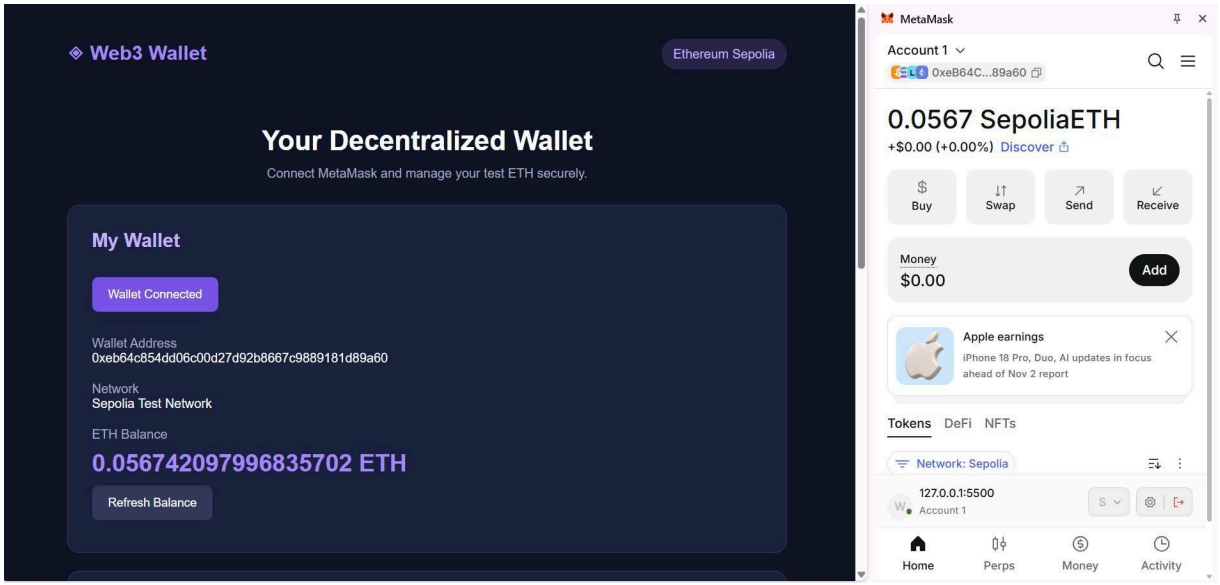
Output & Code:

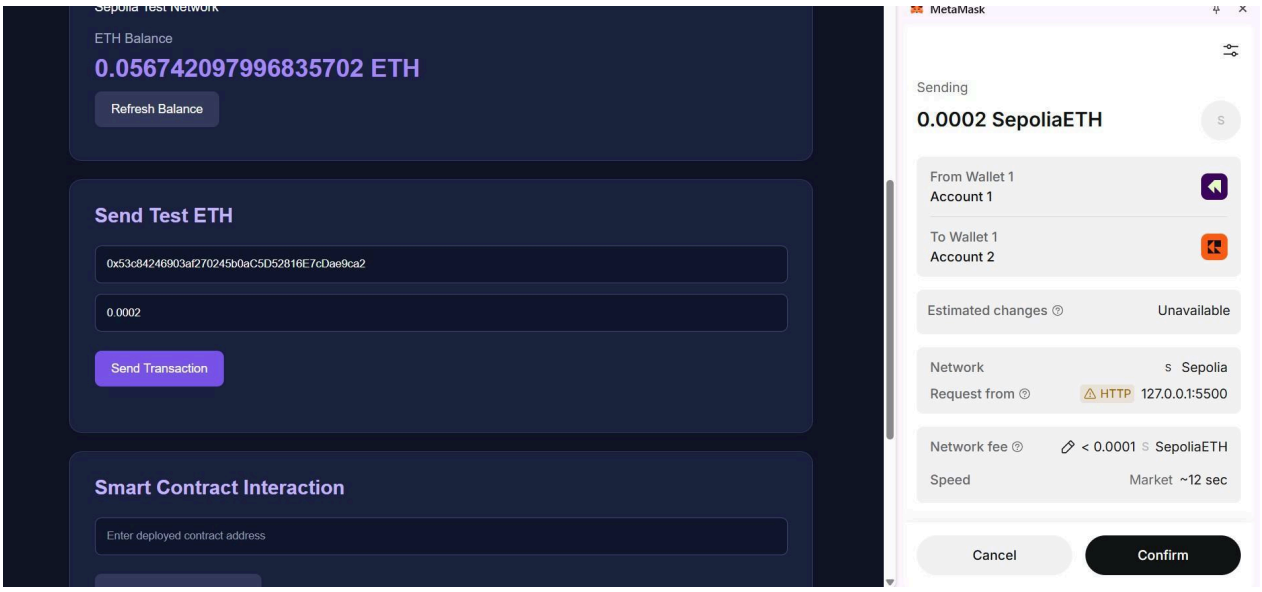


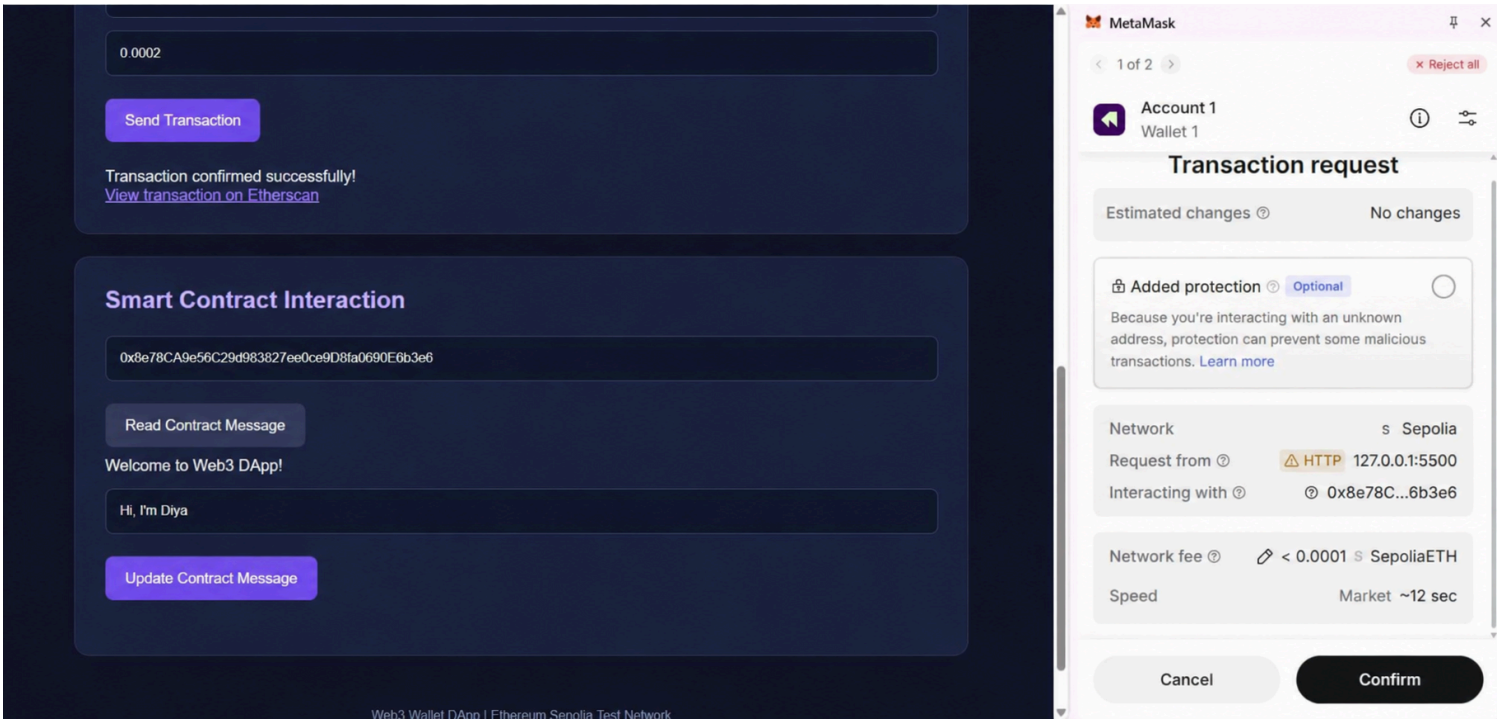


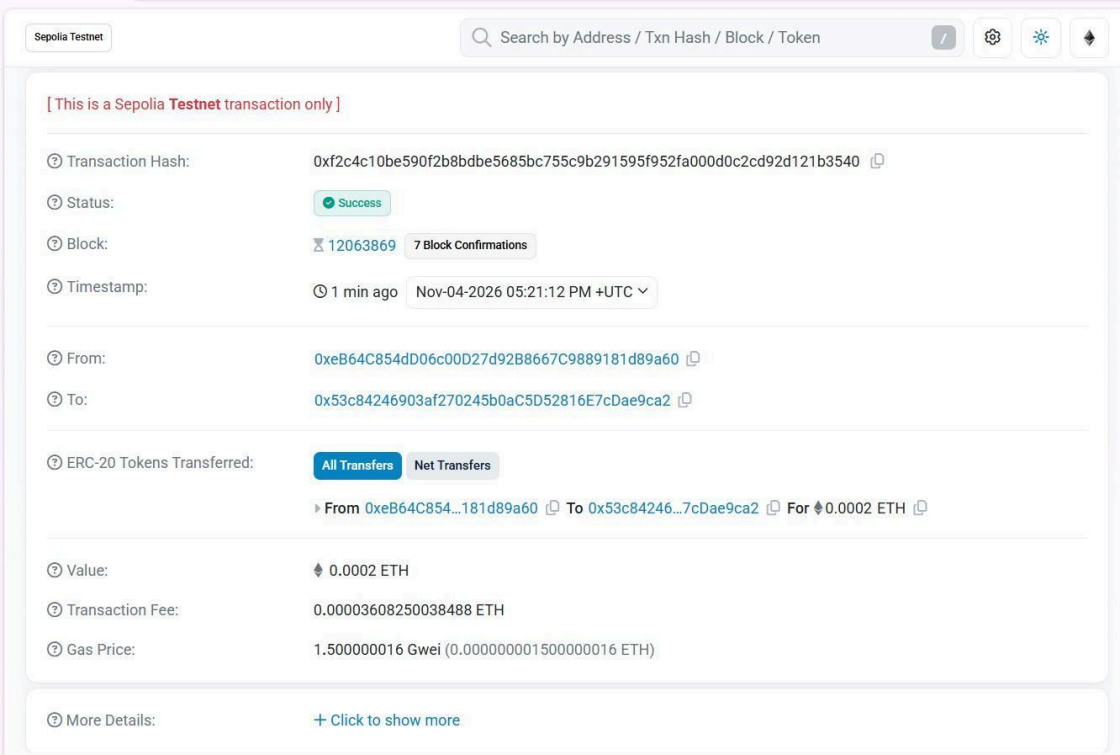

127.0.0.1:5500/index.html
Web3 Wallet
Ethereum Sepolia
Your Decentralized Wallet
Connect MetaMask and manage your test ETH securely.
My Wallet
Connect MetaMask
Send Test ETH
Recipient wallet address
Amount in ETH
Send Transaction
MetaMask
Account 1
0xeB64C...89a60
0.0567 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Money
$0.00
Add
Apple earnings
iPhone 18 Pro, Duo, AI updates in focus ahead of Nov 2 report
Tokens
DeFi
NFTs
Network: Sepolia
SepoliaETH
0.0567 SepoliaETH
Home
Perps
Money
Activity


127.0.0.1:5500
Account 1
Home
Perps
Money
Activity


Web3 Wallet
Ethereum Sepolia
Your Decentralized Wallet
Connect MetaMask and manage your test ETH securely.
My Wallet
Wallet Connected
Wallet Address
0xeb64c854dd06c00d27d92b8667c9889181d89a60
Network
Sepolia Test Network
ETH Balance
0.056742097996835702 ETH
Refresh Balance
MetaMask
Account 1
0xeB64C...89a60
0.0567 SepoliaETH
+$0.00 (+0.00%) Discover
Buy
Swap
Send
Receive
Money
$0.00
Add
Apple earnings
iPhone 18 Pro, Duo, AI updates in focus ahead of Nov 2 report
Tokens
DeFi
NFTs
Network: Sepolia
127.0.0.1:5500
Account 1
Home
Perps
Money
Activity


Sepolia Test Network
ETH Balance
0.056742097996835702 ETH
Refresh Balance
Send Test ETH
0x53c84246903af270245b0aC5D52816E7cDae9ca2
0.0002
Send Transaction
Smart Contract Interaction
Enter deployed contract address
MetaMask
Sending
0.0002 SepoliaETH
From Wallet 1
Account 1
To Wallet 1
Account 2
Estimated changes
Unavailable
Network
Sepolia
Request from
HTTP 127.0.0.1:5500
Network fee
< 0.0001 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm


0.0002
Send Transaction
Transaction confirmed successfully!
View transaction on Etherscan
Smart Contract Interaction
0x8e78CA9e56C29d983827ee0ce9D8fa0690E6b3e6
Read Contract Message
Welcome to Web3 DApp!
Hi, I'm Diya
Update Contract Message
Web3 Wallet DApp | Ethereum Sepolia Test Network
MetaMask
1 of 2
Reject all
Account 1
Wallet 1
Transaction request
Estimated changes
No changes
Added protection
Optional
Because you're interacting with an unknown address, protection can prevent some malicious transactions. Learn more
Network
Sepolia
Request from
HTTP 127.0.0.1:5500
Interacting with
0x8e78C...6b3e6
Network fee
< 0.0001 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm



Conclusion

Thus, MetaMask was successfully configured and connected to the Ethereum Sepolia test network. A simple DApp was created and integrated with MetaMask using Ethers.js. ETH was successfully transferred between two accounts and verified using Sepolia Etherscan. The experiment demonstrated wallet–DApp integration and blockchain transaction processing